

Sardar Vallabhbhai National Institute of Technology Surat
Department of Artificial Intelligence
MID Semester EXAMINATION, B.Tech-III (5th Semester)
Data Science (AI303)

Date: 23-09-2025

MAXIMUM MARKS: 30

Duration: 90 MINUTES

Note: Attempt all questions

S.No	Questions	Marks/CO
Q.1	Define the normal distribution. If the heights of 300 students are normally distributed with a mean of 64.5 inches and a standard deviation of 3.3 inches. How many students have height (a) less than 5 feet b) between 5 feet and 5 feet 9 inches? Also, find the height between 99% of the students' heights. Given that the area under the standard normal curve from $z=0$ to $z=1.36$ is 0.4131, $Z_{0.995} \approx 2.5758$.	[6] (CO-3)
Q.2	Hadoop Distributed File System (HDFS) i. Explain the architecture of HDFS with a neat diagram. Highlight the roles of the NameNode (Master) and DataNodes (Slaves). ii. Explain the concept of heartbeats exchanged between the Resource Manager and Node Managers in Hadoop2.x YARN. Why are heartbeats important for cluster health monitoring? iii. Assume a file of 1 GB is to be stored in a distributed fashion on a Hadoop cluster with 1 Master and 4 Slave machines. Block size = 128 MB, Replication factor = 2. Compute the number of blocks required and illustrate how these blocks can be distributed across the 4 slave machines with replication.	[8] (CO-2)
Q.3	Consider the following two-dimensional data points: (2,1), (3,5), (4,3), (5,6), (6,7), (7,8) (a) Compute the mean vector and the sample covariance matrix. (b) Using Principal Component Analysis (PCA), determine the eigenvalues and eigenvectors of the covariance matrix. Identify the first principal component. (c) Transform the point (2,1) onto the first principal component axis. Report both the principal component score (scalar projection) and the reconstructed point on the PC1 line in the original 2D space. (d) Show how PCA can be derived from the Singular Value Decomposition (SVD) of the centred data matrix. Clearly state the relation between covariance eigenvalues and singular values. (e) Verify your PCA results in parts (b) and (c) using the SVD of the centred data matrix.	[8] (CO-1)
Q.4	A government transport department maintains a dataset of city bus GPS logs to analyse traffic congestion patterns. Each log entry contains the following fields: BusID, RouteID, Timestamp, Latitude, Longitude, Speed The department wants to compute the average speed of buses per route during peak hours (8 AM–11 AM and 5 PM–8 PM) to identify congested routes. (a) Describe how this problem can be solved using the MapReduce framework. Clearly define the Map function, Shuffle & Sort phase, and Reduce function in terms of their input and output key-value pairs. (b) Draw the logical data flow diagram of MapReduce, showing input, intermediate key-value pairs, shuffle/sort, and final output.	[8] (CO-2)